
Understanding VANET (Vehicular Ad Hoc Networks)

 Lecture Note

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Wireless Ad Hoc Network (WANET)

What is a Wireless Ad Hoc Network (WANET)?

A Wireless Ad Hoc Network (WANET) is a decentralized wireless network where nodes communicate without pre-existing infrastructure. Each node acts as both a host and a router, forwarding data for other nodes.

Key Features:

- Infrastructure-less
- Self-configuring
- Multi-hop communication
- Dynamic topology

Summary Table of WANET Types

Type	Mobility	Key Feature	Example Use Case
MANET (Mobile Ad Hoc Network)	Medium	General-purpose mobile network where nodes freely move and form temporary links.	Military units sharing data on the battlefield, search and rescue teams in disaster areas.
VANET (Vehicular Ad Hoc Network)	High (vehicular)	High-speed node mobility with short connection windows; supports V2V and V2I communication.	Traffic safety systems, collision avoidance, intelligent transport systems (ITS).
WSN (Wireless Sensor Network)	Low or static	Composed of low-power sensor nodes focused on data collection and efficient communication.	Forest fire detection, environmental monitoring, industrial automation.
WMN (Wireless Mesh Network)	Low (for routers)	Fixed backbone of mesh routers with self-healing multi-hop communication; supports client mobility.	City-wide public Wi-Fi, rural broadband internet distribution.
FANET (Flying Ad Hoc Network)	Very high (3D)	Supports dynamic 3D topologies involving UAVs with aerial communication links.	Aerial surveillance, disaster zone mapping, drone swarm coordination.
UAN (Underwater Ad Hoc Network)	Slow, underwater	Uses acoustic signaling for underwater communication with long propagation delays.	Submarine communication, oceanographic data collection, underwater exploration.
SPAN (Space-based Ad Hoc Network)	Orbital/space	Delay-tolerant networking among satellites or space probes over large interplanetary distances.	Satellite constellations for Earth observation, Mars rover communication.

Table 1: Summary of WANET Types

Classification of Wireless Ad Hoc Networks (WANETs)

2. Wireless Ad Hoc Networks can be classified based on mobility, node type, infrastructure support, and application domain:

classified by various criteria:

- **By Node Mobility:**
 - Static Nodes (e.g., WSN)
 - Mobile Nodes: MANET, VANET, FANET, SPAN
- **By Node Type:**
 - Homogeneous
 - Heterogeneous (mix of UAVs, sensors, routers)
- **By Infrastructure Support:**
 - Infrastructure-less
 - Hybrid (with RSUs/gateways)
- **By Application Domain:**
 - Military, Disaster Relief, Smart Environments, PANs
- **By Routing Protocol Design:**
 - Proactive (e.g., OLSR)
 - Reactive (e.g., AODV)
 - Hybrid (e.g., ZRP)
 - Geographic (e.g., GPSR)

VANET (Vehicular Ad Hoc Network)

What is VANET (Vehicular Ad Hoc Network)?

A VANET is a specialized type of Mobile Ad Hoc Network (MANET) where vehicles act as nodes. It supports communication between:

- Vehicle-to-Vehicle (V2V)
- Vehicle-to-Infrastructure (V2I)
- Vehicle-to-Network (V2N) (in advanced settings)

VANETs are a high-mobility, safety-critical subclass of wireless ad hoc networks with:

- Flat or layered architecture depending on deployment
- Multi-channel MAC using IEEE 802.11p/WAVE standards
- Predominantly single-radio setups with synchronized channel switching
- Ability to evolve into multi-radio, out-of-band hybrid systems for smart infrastructure.

Device Hierarchy in VANET

Type	Description	Example Use Case
Flat	All vehicles are peers (equal roles)	Safety messaging, decentralized alerts
Layered	Roles vary: Vehicles, RSUs, Cloud backend	Infrastructure-assisted smart traffic

VANETs are mostly flat but can adopt a layered hierarchy in hybrid or smart city environments.

Communication Standards

The two major **vehicular wireless communication standards**:

1. **IEEE 802.11p**
2. **Cellular V2X (C-V2X).**

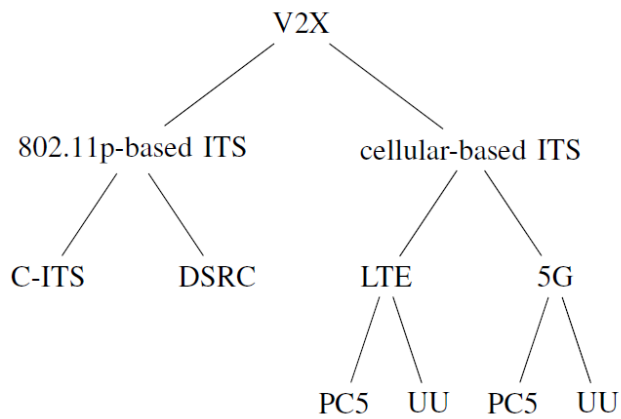


Figure 1: Taxonomy of used V2X communications

IEEE 802.11p vs. C-V2X in VANET Communication

Feature	IEEE 802.11p (DSRC/WAVE)	C-V2X (Cellular-V2X)
Origin / Standard	IEEE (part of IEEE 802.11 family)	3GPP (Release 14+, based on LTE and 5G NR)
Full Name	Wireless Access in Vehicular Environments (WAVE)	Cellular Vehicle-to-Everything (C-V2X)
Technology Basis	Wi-Fi based (ad hoc MAC, 802.11)	Cellular-based (LTE/5G NR sidelink)
Frequency Band	5.850–5.925 GHz (USA)	5.9 GHz (for V2X), shared with 4G/5G networks
Communication Modes	Direct (V2V, V2I) without infrastructure	PC5 (Direct): Vehicle-to-Vehicle (V2V) or Vehicle-to-RSU communication over sidelink — no cellular infrastructure required. Or, Uu (Indirect): Communication via base station and core network — infrastructure-dependent (used in V2N, V2I via cellular).
Latency (V2V)	~10 ms (real-time)	~5–10 ms (comparable or better in newer releases)
Range	~300–1000 meters	~300–1000 meters
MAC Layer Access	CSMA/CA (contention-based)	Scheduled or semi-persistent resource allocation
Broadcast Efficiency	Degrades under high load (collision-prone)	More reliable in dense environments due to scheduling
Scalability	Limited in high-density scenarios	Better support for scalability and QoS in dense areas
Deployment Status	Widely tested, deployed in some countries	Rapidly gaining adoption globally with 4G/5G integration
Future Outlook	Likely to coexist or be replaced by C-V2X in many regions	Strong momentum, especially with 5G-V2X support

Table 4: IEEE 802.11p vs. C-V2X

Key Differences Explained

1. Communication Model

- **IEEE 802.11p**: Purely ad hoc. Vehicles broadcast messages without central scheduling. Good for quick deployment but suffers from collisions in dense networks.
- **C-V2X**: Can operate both **directly (PC5 interface)** and via **cellular towers (Uu interface)**. More flexible and reliable under load.

2. Medium Access Control

- **IEEE 802.11p** uses **CSMA/CA based QoS enabled EDCA**, meaning it contends for the channel, which causes **hidden/exposed terminal problems**.
- **C-V2X** uses **SPS (Semi-Persistent Scheduling)** or **network-controlled allocation**, offering **collision-free** access and **QoS assurance**.
 - **SPS (Semi-Persistent Scheduling)** is a **resource allocation technique** used in **C-V2X**, particularly over the **PC5 (sidelink)** interface.
 - **SPS** allows a vehicle (UE) to **reserve radio resources periodically** (e.g., every 100 ms) for a fixed duration **without requesting resources each time**, reducing control signaling overhead.
 -  **Key Features:**
 -  **Semi-persistent**: Resources are reserved for multiple transmissions, then reselected.
 -  **Low Latency**: Reduces delay by avoiding frequent scheduling requests.
 -  **Efficient**: Minimizes signaling and supports high reliability in broadcast scenarios like **BSM**.
 -  **Distributed**: Each UE independently selects resources based on sensing.

3. Scalability and Performance

- **802.11p** degrades significantly in urban or congested settings.
- **C-V2X** handles dense environments better and can support **more simultaneous users**.

4. Deployment and Ecosystem

- **IEEE 802.11p** has been **used in pilot deployments** (e.g., US DOT Safety Pilot), especially where **DSRC infrastructure** exists.
- **C-V2X** is being **embraced globally**, especially by telecom providers and **automakers (e.g., Ford, Audi, BMW)** due to its **forward compatibility with 5G**.

IEEE 802.11p was the first standard for vehicular communication with simple ad hoc design, but its scalability limitations have motivated a shift toward C-V2X, which offers better reliability, QoS, and integration with future 5G infrastructure.

Vehicular Wireless Communication Using IEEE 802.11p

Vehicular wireless communication using IEEE 802.11p is being planned globally. [Table 1](#) provides a high-level comparison of aspects in Japan, US, and EU.

	Japan	USA	Europe
Standard / Committee	ITS-Forum	IEEE802.11p/1609.x	CEN/ETSI EN302 663
Frequency range	755 – 765 MHz	5850 – 5925 MHz	5855 – 5925 MHz
No. of Channels	One 10 MHz channel	Seven 10 MHz channels (Two 20 MHz channels formed by combining 10 MHz channels)	Seven 10 MHz channels
Modulation	OFDM		
Data rate per channel	3 -18Mbit/s	3 -27Mbit/s	3 - 27Mbit/s
Output power	20 dBm (Antenna input)	23 - 33 dBm (EIRP) ¹	23-33 dBm (EIRP)
Communication	One direction multicasting service (broadcast without ACK)	One direction multicasting service, One to Multi communication, Simplex communication (broadcast without ACK, multicast, unicast with ACK)	
Upper protocol	ARIB STD-T109	WAVE (IEEE 1609) / TCP/IP	ETSI EN 302 665 (incl. e.g. GeoNetworking) TCP/UDP/IP

Table 1: High level global overview of ITS.

IEEE 802.11p (DSRC/WAVE) for Vehicular Communication

IEEE 802.11p is a wireless communication standard tailored for vehicular environments. It is part of the IEEE 802.11 family and forms the foundation for Dedicated Short-Range Communications (DSRC) and Wireless Access in Vehicular Environments (WAVE). It enables Vehicle-to-Vehicle (V2V) and Vehicle-to-Infrastructure (V2I) communication without relying on cellular infrastructure.

Architecture and Stack Overview

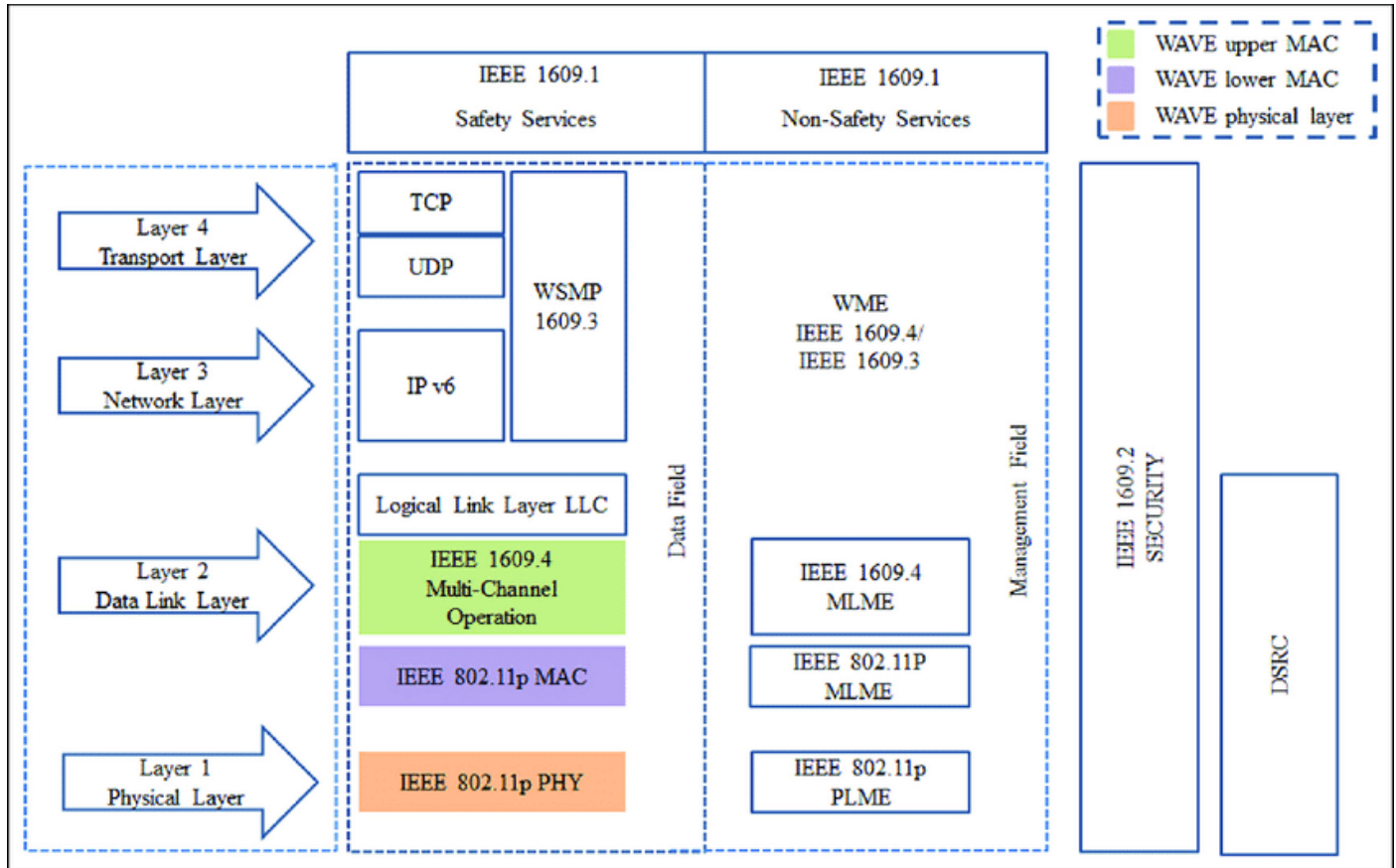


Figure 1: WAVE protocol stack.

IEEE 802.11p operates in the 5.850–5.925 GHz band (in the US) and supports a 10 MHz channel bandwidth to improve signal robustness in high-mobility environments. Operates in the licensed ITS band at 5.9 GHz designed for high-speed vehicular environments. It fits within the WAVE protocol stack, which includes:

- **IEEE 1609.1:** Application & resource management
- **IEEE 1609.2:** Security services (authentication, message integrity).
- **IEEE 1609.3:** Network services and message format (e.g., WSMP(Wave Short Message Protocol)).
- **IEEE 1609.4:** Multi-channel operation (CCH/SCH switching).
- **IEEE 802.11p:** PHY and MAC layer enhancements for vehicular use.

Channel Structure

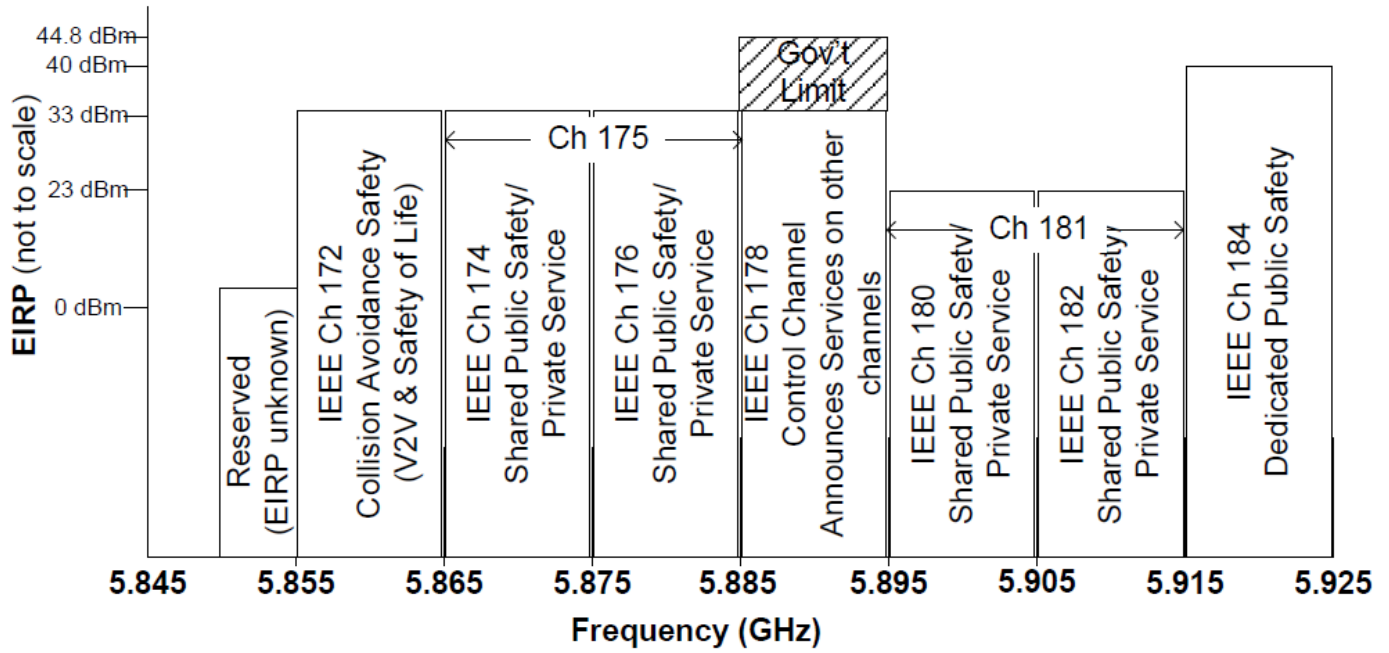


Figure 3-1: US ITS/DSRC channel plan

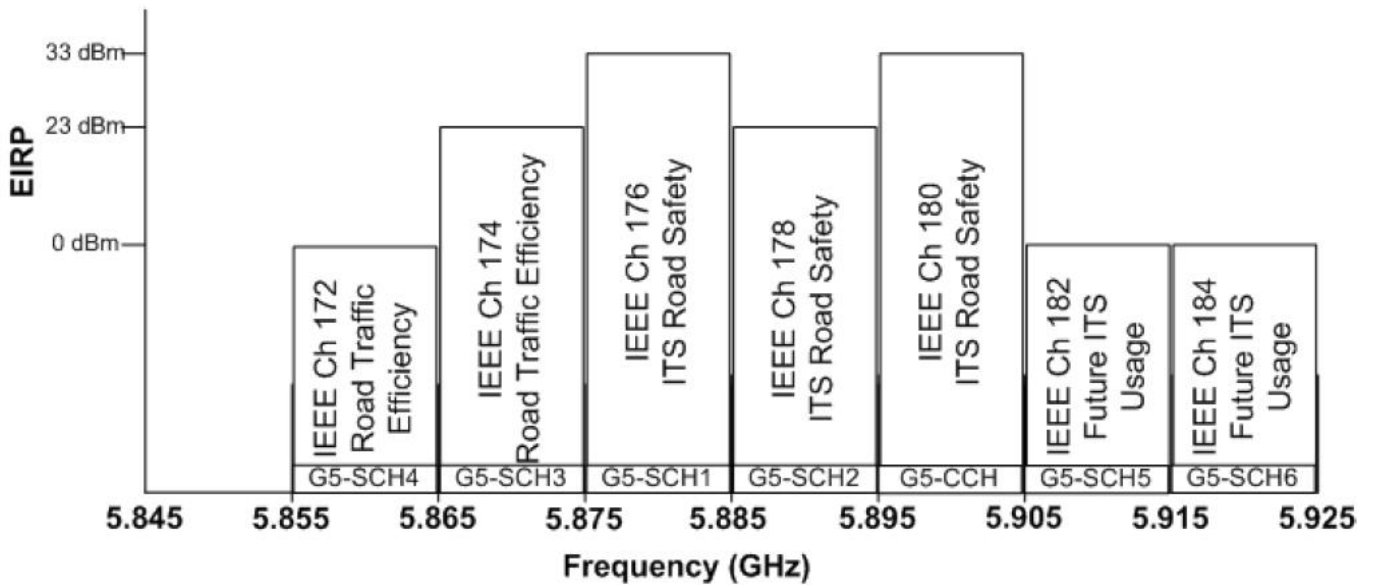


Figure 3-2: Europe ITS Channel Plan

US ITS Channel Structure

The **75 MHz licensed ITS band** (5.850–5.925 GHz) in the United States is divided into **seven 10 MHz-wide channels** allocated as follows:

Channel	Frequency (GHz)	Designation	Usage Category
CH 172	5.855–5.865	Collision Avoidance (V2V, Safety-of-Life)	Dedicated Safety
CH 174	5.865–5.875	Shared Public/Private Safety	Safety/Service
CH 176	5.875–5.885	Shared Public/Private Safety	Safety/Service
CH 178	5.885–5.895	Control Channel (CCH)	Broadcasts WSAs (Service Access)
CH 180	5.895–5.905	Shared Public/Private Safety	Safety/Service
CH 182	5.905–5.915	Shared Public/Private Safety	Safety/Service
CH 184	5.915–5.925	High-Power Public Safety	Extended Safety (e.g., intersections)

Channel Switching Timing

- Devices **alternate between CCH (CH 178) and SCHs** using **synchronized 100 ms intervals**:
 - 50 ms** on Control Channel (CCH)
 - 50 ms** on a selected Service Channel (SCH)

Channel Roles

- Control Channel (CCH - CH 178):**
 - Used to broadcast **WAVE Service Advertisements (WSAs)** and safety-critical messages (e.g., BSM).
- Service Channels (SCHs):**
 - Used for **application data**, including infotainment, cooperative driving data, non-safety updates.
- CH 172 & CH 184:**
 - Strictly reserved for safety-critical applications.** CH 184 allows **higher power transmission (up to 44.8 dBm)** for government/public safety.

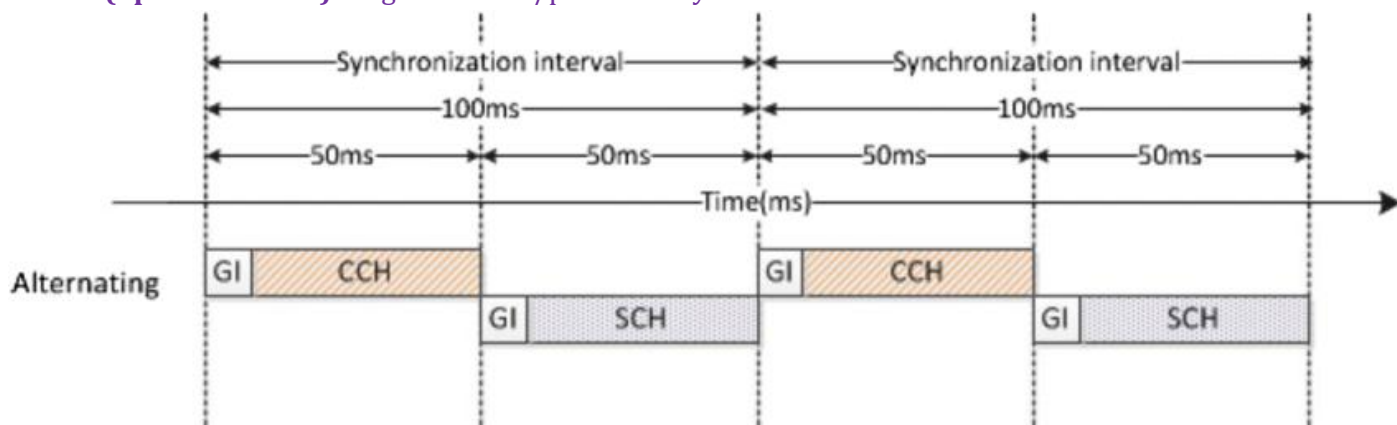


Figure 4: IEEE Std. 1609.4 alternating mode switching time

Applications of IEEE 802.11p

- **Emergency Electronic Brake Light (EEBL):** Alert to following vehicles when a leading vehicle brakes suddenly.
- **Intersection Movement Assist (IMA):** Warning to vehicles approaching intersections with potential collision risk.
- **Forward Collision Warning (FCW):** Notifies a driver of an impending crash with a vehicle ahead in the same lane.
- **Cooperative Adaptive Cruise Control (C-ACC):** Vehicles share speed and acceleration info to coordinate smoother platooning.
- **RSU-Based Traffic Alerts:** Roadside units broadcast localized updates such as traffic conditions, hazards, or speed limits.

Safety vs. Non-Safety Applications in IEEE 802.11p

Safety Applications

- **Purpose:** Prevent accidents and enhance road safety through timely alerts.
- **Examples:**
 - Forward Collision Warning
 - Emergency Brake Light
 - Intersection Movement Assist
- **Primary Message Type:**
 - **BSM (Basic Safety Message)** – Periodically broadcast vehicle status (position, speed, heading).
- **Channel Used:**
 - **Control Channel (CCH)** or **Safety Channel (SfCH)** (typically centered at 5.9 GHz in the U.S.)
- **Access Method:**
 - Sent without connection setup, often broadcast to all nearby vehicles.

Non-Safety Applications

- **Purpose:** Enhance driver convenience and infotainment; not mission-critical.
- **Examples:**
 - Toll collection
 - Map updates
 - Internet access
- **Primary Message Types:**
 - **WSA (WAVE Service Advertisement)** – Broadcasts the availability of a service on a specific channel.
 - **WSM (WAVE Short Message)** – Payload message sent after WSA is received/accepted.
- **Channel Used:**
 - **Service Channels (SCHs)** (e.g., 5.860 GHz–5.880 GHz, 5.910 GHz–5.920 GHz)
- **Access Method:**
 - Follows WSA → SCH handover → WSM exchange.

Control Channel vs. Service Channels

Channel Type	Purpose	Used For
CCH	Broadcast safety info (BSM, WSA)	Safety alerts, service advertisements
SCH	Non-safety service data (WSM)	Infotainment, commercial apps
SfCH	Safety-focused SCH in some regions	High-priority safety data (EU/US)

Message Flow Summary

1. Safety Flow:

Vehicle → BSM → CCH (Broadcast regularly)

2. Non-Safety Flow:

RSU → WSA (on CCH) → Vehicle switches to SCH → Vehicle ↔ RSU via WSM

Applications that use BSM as their core data source:

- Emergency Electronic Brake Light (EEBL)**
 - Uses the **brake status and speed** field from received BSMs of nearby vehicles.
- Intersection Movement Assist (IMA)**
 - Monitors **position, speed, and heading** from BSMs to assess intersection conflicts.
- Forward Collision Warning (FCW)**
 - Requires **distance, speed, and deceleration** info from the BSMs of leading vehicles.
- Cooperative Adaptive Cruise Control (C-ACC)**
 - Depends on **periodic BSMs** exchanged between platooning vehicles for real-time coordination.

Application that does not use BSM directly:

- RSU-Based Traffic Alerts**
 - These messages are typically delivered using **MAP, SPaT, or TIM messages** (not BSM).
 - RSUs may **receive BSMs** for awareness but broadcast other message types for advisories.

Single Channel vs Multi-Channel in VANET

Feature	Single Channel	Multi-Channel
Complexity	Low	Higher (requires coordination)
Throughput	Limited	Improved
Collision	More likely	Reduced via channel separation
Usage in VANET	Possible, but less efficient	Standard in VANET (WAVE/DSRC)

Single Radio vs Multi-Radio in VANET

Feature	Single Radio (Typical)	Multi-Radio (Advanced)
Channel Access	Alternates between CCH and SCH	Simultaneous (CCH and SCH)
Hardware Cost	Low	Higher
Complexity	Moderate	High (coordination needed)
Use Case	Standard VANET	Smart cities, high-density areas

Most VANETs use a single radio that switches between control and service channels, but multi-radio setups are used in advanced deployments.

IEEE 802.11p Medium Access Control (MAC) Overview

IEEE 802.11p adopts **Enhanced Distributed Channel Access (EDCA)**, a QoS-aware extension of CSMA/CA, from the IEEE 802.11e standard. It is optimized for the broadcast-heavy and latency-sensitive nature of vehicular environments:

-  **Priority Queues:** Uses 4 Access Categories (ACs) — Voice, Video, Best Effort, Background — with different contention parameters.
-  **Fast Access:** Higher-priority messages (e.g., safety alerts) get shorter contention windows and inter-frame spaces.
-  **Broadcast-Focused:** Designed for frequent, short, one-hop safety messages where acknowledgments are not feasible.
-  **No RTS/CTS:** Omits request-to-send/clear-to-send exchange to reduce delay in fast-moving networks.

Advantages and Limitations

Advantages:

- Infrastructure-less direct communication (suitable for V2V).
- Mature, standardized and tested.

Limitations:

- Collision-prone under dense traffic due to CSMA/CA.
- No QoS guarantees.
- Limited scalability.
- Risk of hidden terminal problems.

IEEE 802.11p Physical Layer (PHY) Overview

IEEE 802.11p adopts the OFDM-based PHY defined in **802.11a**, but operates primarily in **10 MHz bandwidth** mode (half clocked), suitable for high-mobility vehicular environments. Key characteristics include:

- **Bandwidth Options:** 5 MHz, 10 MHz (used in 802.11p), 20 MHz (used in 802.11a).
- **Subcarriers:** 64-point FFT with 52 subcarriers (48 data + 4 pilot).
- **Symbol Duration:** Extended in narrower bands (e.g., 8 μ s for 10 MHz).
- **Modulations:** BPSK, QPSK, 16-QAM, 64-QAM with coding rates 1/2, 2/3, 3/4.
- **Preamble & Guard Times:** Doubled for 10 MHz vs 20 MHz for better delay spread handling.

Stricter Requirements:

- **Tighter spectrum masks** and **adjacent/non-adjacent channel rejection** are enforced in 802.11p to mitigate **cross-channel interference**, especially in multi-channel ITS setups.

Why It Matters:

- These changes enhance **communication reliability** under interference and **long-range V2X scenarios**.

Table 2 gives a comparison of key OFDM parameters as a function of the bandwidth.

Parameters	20 MHz Bandwidth	10 MHz Bandwidth	5 MHz Bandwidth
Bit rate (Mbit/s)	6, 9, 12, 18, 24, 36, 48, 54	3, 4.5, 6, 9, 12, 18, 24, 27	1.5, 2.25, 3, 4.5, 6, 9, 12, 13.5
Modulation mode	BPSK, QPSK, 16QAM, 64QAM	BPSK, QPSK, 16QAM, 64QAM	BPSK, QPSK, 16QAM, 64QAM
Code rate	1/2, 2/3, 3/4	1/2, 2/3, 3/4	1/2, 2/3, 3/4
Number of subcarriers	52	52	52
Symbol duration	4 μ s	8 μ s	16 μ s
Guard time	0.8 μ s	1.6 μ s	3.2 μ s
FFT period	3.2 μ s	6.4 μ s	12.8 μ s
Preamble duration	16 μ s	32 μ s	64 μ s
Subcarrier spacing	312.5 kHz	156.25 kHz	78.125 kHz

Table 2: 802.11 OFDM PHY Parameters as a function of channel bandwidth

Deployment and Adoption

IEEE 802.11p has been piloted and deployed in the US (e.g., Safety Pilot Model Deployment by USDOT), Europe (under ETSI ITS-G5), and Japan. However, adoption is slowing due to the emergence of cellular alternatives like C-V2X.

IEEE 802.11p was the first wireless standard designed specifically for vehicular communication. Despite its drawbacks in dense environments, it laid the groundwork for V2X systems and still plays a critical role in certain deployments.